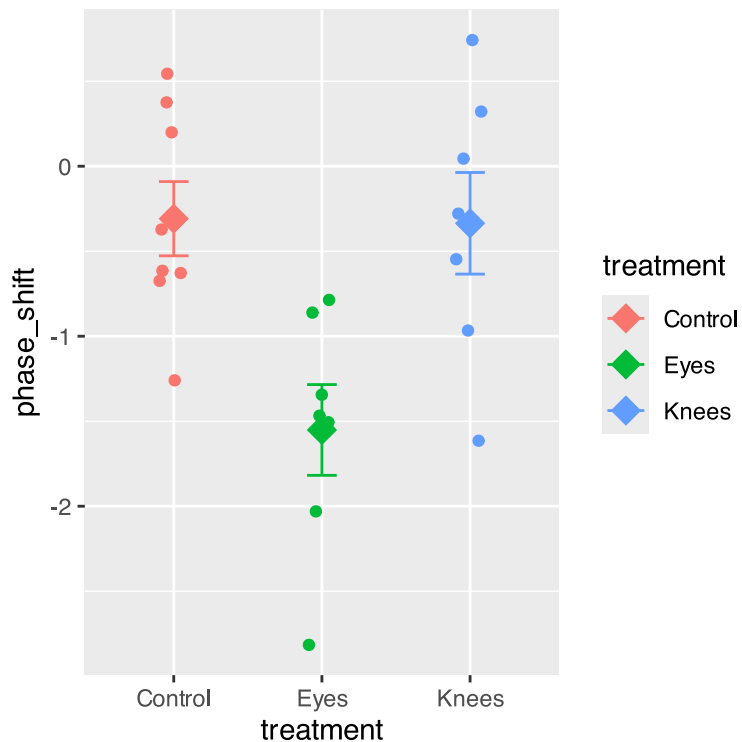


Lecture 11 - ANOVA in class

Bill Perry

```
# A tibble: 22 × 2
  treatment phase_shift
<chr>      <dbl>
1 Control    0.53
2 Control    0.36
3 Control    0.2
4 Control   -0.37
5 Control   -0.6
6 Control   -0.64
7 Control   -0.68
8 Control   -1.27
9 Knees      0.73
10 Knees     0.31
# i 12 more rows
```

```
# Plot the data
c_plot <- ggplot(c_df, aes(x = treatment, y = phase_shift, color = treatment)) +
  geom_point(position = position_jitter(width=0.1)) +
  stat_summary(fun = mean, geom = "point", size = 5, shape = 18) +
  stat_summary(fun.data = "mean_se", geom = "errorbar", width = 0.2)
c_plot
```

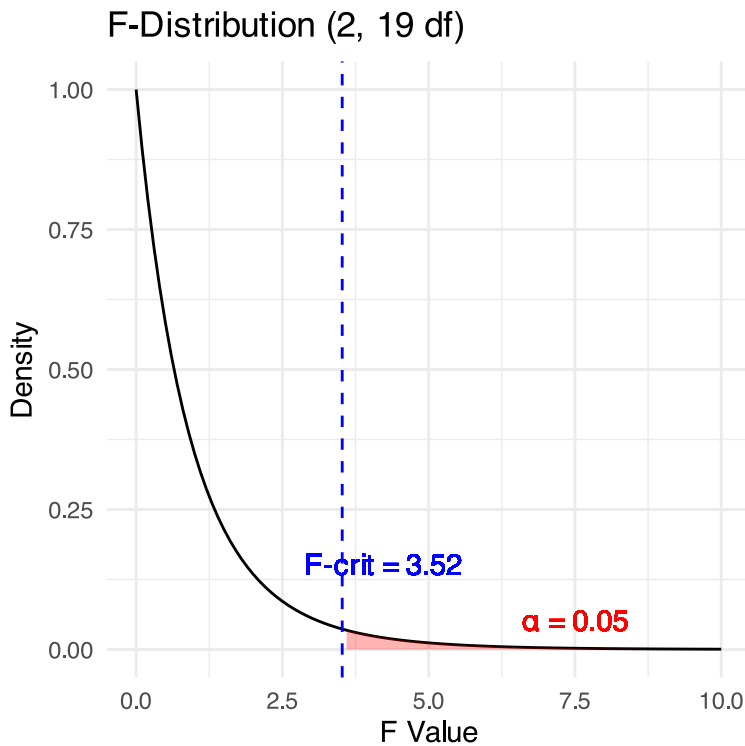


Anova Table (Type III tests)

Response: phase_shift

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	0.7626	1	1.5389	0.229877
treatment	7.2245	2	7.2894	0.004472 **
Residuals	9.4153	19		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



ANOVA-Assumptions and Diagnostics

ANOVA has the same assumptions as the two-sample t-test, but applied to all k groups:

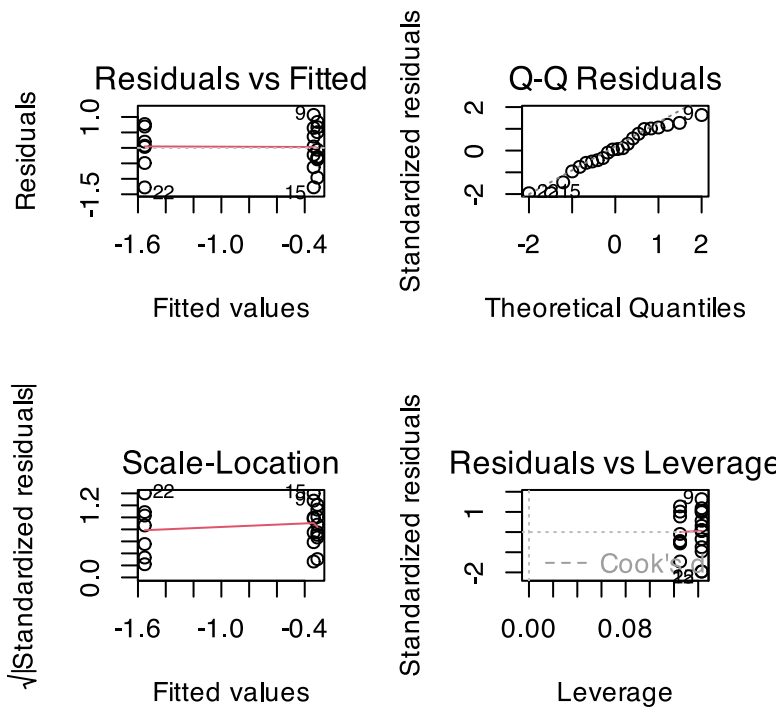
1. **Random samples** from corresponding populations
2. **Normality**: Y values are normally distributed in each population
3. **Homogeneity of variance**: variance is the same in all populations
4. **Independence**: observations are independent

Checking assumptions:

- Normality: Q-Q plots, histogram of residuals, Shapiro-Wilk test
- Homogeneity: plot residuals vs. predicted values or x-values
- Independence: examine experimental design

Lecture 12: ANOVA diagnostics

This is the default output of base R



ANOVA Diagnostics

Levene's test of homogeneity of variance

- Null Hypothesis is that they are homogeneous
- So you want a non significant result here

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.1586 0.8545
      19
```

Lecture 12: ANOVA Diagnostics

Shapiro-Wilk Normality Test

- Null Hypothesis is that they are normally distributed
- So you want a **non significant** result here

Shapiro-Wilk normality test

```
data: residuals(model_aov)
W = 0.95893, p-value = 0.468
```

Lecture 12: ANOVA Post-Hoc Testing

When ANOVA rejects H_0 , we need to determine which groups differ.

Example: Using Tukey's HSD to compare all pairs of treatments in the circadian rhythm data.

contrast	estimate	SE	df	t.ratio	p.value
Control - Eyes	1.243	0.364	19	3.411	0.0088
Control - Knees	0.027	0.364	19	0.074	0.9998
Eyes - Knees	-1.216	0.376	19	-3.231	0.0131

P value adjustment: sidak method for 3 tests

```
emmeans_df <- emmeans(model_aov, "treatment")
emmeans_df
```

treatment	emmean	SE	df	lower.CL	upper.CL
Control	-0.309	0.249	19	-0.830	0.212
Eyes	-1.551	0.266	19	-2.108	-0.995
Knees	-0.336	0.266	19	-0.893	0.221

Confidence level used: 0.95

```
pairwise_comparisons <- pairs(emmeans_df, adjust = "sidak")
pairwise_comparisons
```

contrast	estimate	SE	df	t.ratio	p.value
Control - Eyes	1.243	0.364	19	3.411	0.0088
Control - Knees	0.027	0.364	19	0.074	0.9998
Eyes - Knees	-1.216	0.376	19	-3.231	0.0131

P value adjustment: sidak method for 3 tests

```
# Letters to indicate groups with similar means
letter_groups <- multcomp::cld(emmeans_df, Letters = letters, adjust = "sidak")
letter_groups
```

treatment	emmean	SE	df	lower.CL	upper.CL	.group
Eyes	-1.551	0.266	19	-2.25	-0.855	a
Knees	-0.336	0.266	19	-1.03	0.361	b
Control	-0.309	0.249	19	-0.96	0.343	b

Confidence level used: 0.95

Conf-level adjustment: sidak method for 3 estimates

P value adjustment: sidak method for 3 tests

significance level used: alpha = 0.05

NOTE: If two or more means share the same grouping symbol,
then we cannot show them to be different.
But we also did not show them to be the same.

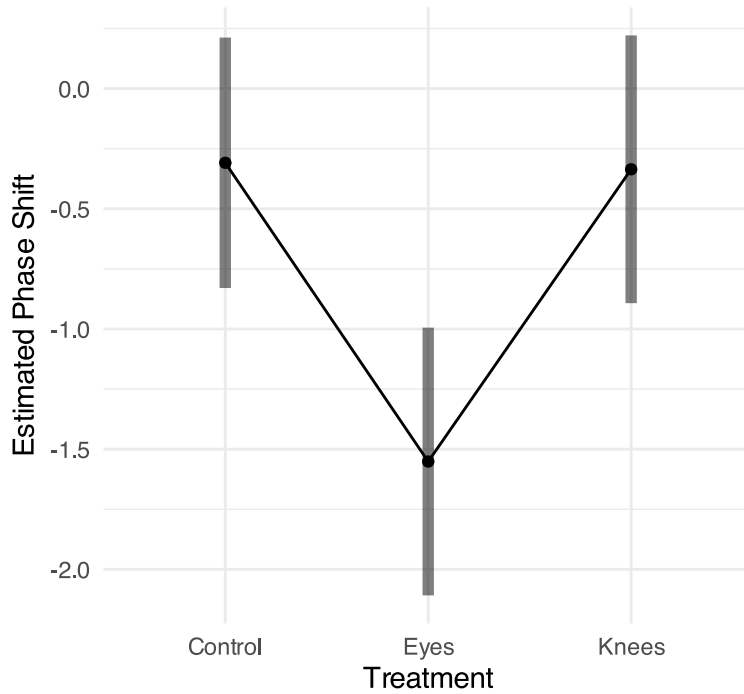
Can do in one go as well

treatment	emmean	SE	df	lower.CL	upper.CL	.group
Eyes	-1.551	0.266	19	-2.25	-0.855	a
Knees	-0.336	0.266	19	-1.03	0.361	b
Control	-0.309	0.249	19	-0.96	0.343	b

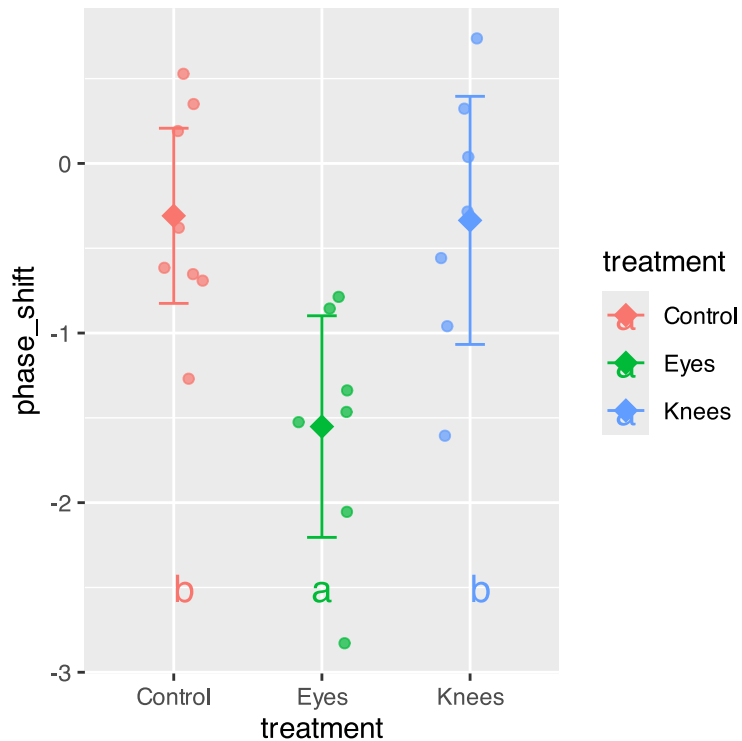
Confidence level used: 0.95
Conf-level adjustment: sidak method for 3 estimates
P value adjustment: sidak method for 3 tests
significance level used: $\alpha = 0.05$
NOTE: If two or more means share the same grouping symbol,
then we cannot show them to be different.
But we also did not show them to be the same.

Lecture 12: ANOVA Post-Hoc Testing

Estimated Marginal Means with 95% CIs



Lecture 12: ANOVA Post-Hoc Testing



Lecture 12: ANOVA Reporting results

Formal scientific writing example:

“The effect of light treatment on circadian rhythm phase shift was analyzed using a one-way ANOVA. There was a significant effect of treatment on phase shift ($F(2, 19) = 7.29$, $p = 0.004$, $\eta^2 = 0.43$). Post-hoc comparisons using Tukey’s HSD test indicated that the mean phase shift for the Eyes treatment ($M = -1.55$ h, $SD = 0.71$) was significantly different from both the Control treatment ($M = -0.31$ h, $SD = 0.62$) and the Knees treatment ($M = -0.34$ h, $SD = 0.79$). However, the Control and Knees treatments did not significantly differ from each other. These results suggest that light exposure to the eyes, but not to the knees, impacts circadian rhythm phase shifts.”